Nifty Intraday Options Strategies Backtest Report

Qode Advisors LLP — Quant Research Analyst Assessment

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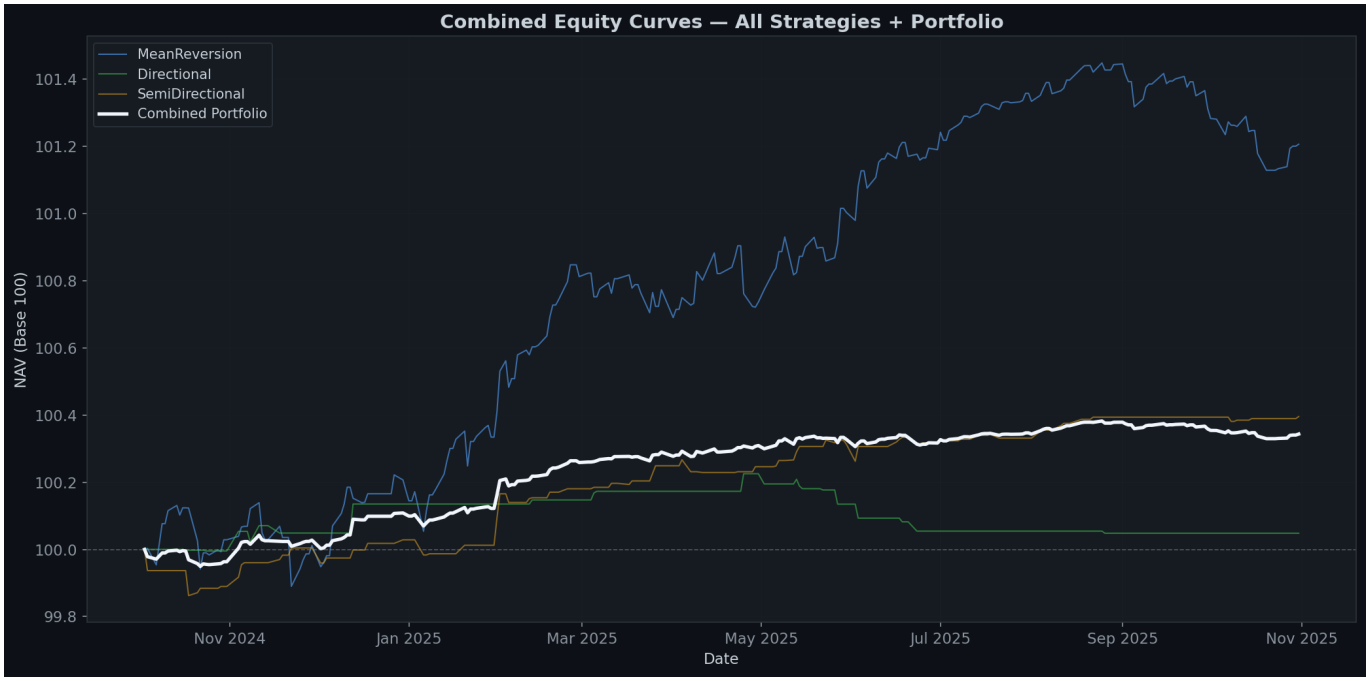
Total Strategies	3
Data Period	Oct 2024 — Mar 2026
In-Sample	Oct 2024 — Oct 2025
Out-of-Sample	Nov 2025 — Mar 2026
Combined Calmar	6.0894
Combined Sharpe	-48.1715
Total Net PnL	Rs. 34,333

1. Executive Summary

This report presents the backtest results of three intraday Nifty options trading strategies designed to capture different market regimes: mean reversion, directional momentum, and semi-directional bias. Each strategy is backtested on 1-minute OHLCV options data from October 2024 to October 2025 (in-sample) and validated on November 2025 to March 2026 (out-of-sample). The combined portfolio is optimized for maximum Calmar ratio.

Performance Summary — In-Sample

Strategy	CAGR	Max DD	Sharpe	Calmar	Win Rate	Net PnL
MeanReversion	1.1%	-0.3%	-9.89	3.56	52.2%	Rs.40,196
Directional	0.0%	-0.2%	-46.10	0.25	3.0%	Rs.1,604
SemiDirectional	0.4%	-0.1%	-33.11	2.67	20.2%	Rs.13,180
Combined	0.3%	-0.1%	-48.17	6.09	54.1%	Rs.34,333
Combined	**0.3%**	**-0.1%**	**-48.17**	**6.09**	**54.1%**	**Rs.34,333**



2. Data & Methodology

2.1 Data Source

1-minute OHLCV data for NIFTY options sourced from Google Drive. The dataset covers October 2024 to March 2026 (~17 months) with options organized by weekly expiry dates. Each expiry folder contains CSV files for individual strike-type combinations.

2.2 Data Processing

Raw CSV files were parsed, standardized, and filtered to market hours (09:15-15:29). Synthetic Nifty spot prices were derived using put-call parity at each timestamp. Data was split into in-sample (Oct 2024 - Oct 2025) and out-of-sample (Nov 2025 - Mar 2026) periods.

2.3 Transaction Cost Model

Component	Rate
Brokerage	0 (Zero transaction costs assumed)
STT	0
Exchange charges	0
GST	0
Slippage	0

2.4 Execution Timing

Phase	Time (seconds)
Data Loading	21.73s
In-Sample Strategies	324.74s
Portfolio Construction	0.25s
Out-of-Sample	74.92s

Strategy 1: Mean Reversion (Short Straddle)

Concept: Sells ATM straddle at 09:20, profits from time decay and range-bound markets.

Entry: Sell 1 ATM CE + 1 ATM PE at 09:20

Exit: Combined SL: 1.5x entry premium. Target: 0.5x entry. Time exit: 15:15.

Edge: Theta decay provides a statistical edge. Most days, Nifty stays within the straddle range.

Weakness: Vulnerable to large intraday moves and gap openings. Loses big on trending days.

Performance Metrics (In-Sample)

Metric	Value
CAGR	1.12%
Max Drawdown	-0.31%
Sharpe Ratio	-9.8888
Sortino Ratio	-12.0932
Calmar Ratio	3.5579
Win Rate	52.24%
Profit Factor	1.4292
Payoff Ratio	1.3067
Max Consecutive Wins	5
Max Consecutive Losses	5
Recovery Factor	3.7751
Expectancy	Rs.150
Annual Volatility	0.59%
Total Net PnL	Rs.40,196
Total Trading Days	268

Strategy 2: Directional (Momentum Breakout)

Concept: Buys OTM options in the direction of the first 30-minute breakout.

Entry: Buy OTM CE (ATM+150) if Nifty up >0.5% by 09:45, or OTM PE (ATM-150) if down >0.5%.

Exit: SL: 40% of premium. Target: 100% gain. Trailing SL at 50%/80% profit levels.

Edge: Captures strong intraday trends with limited downside (premium paid).

Weakness: Many false breakouts. Low win rate compensated by large winners.

Performance Metrics (In-Sample)

Metric	Value
CAGR	0.04%
Max Drawdown	-0.18%
Sharpe Ratio	-46.1013
Sortino Ratio	-79.4492
Calmar Ratio	0.2531
Win Rate	2.99%
Profit Factor	1.1928
Payoff Ratio	38.7681
Max Consecutive Wins	2
Max Consecutive Losses	119
Recovery Factor	0.2717
Expectancy	Rs.6
Annual Volatility	0.15%
Total Net PnL	Rs.1,604
Total Trading Days	268

Strategy 3: Semi-Directional (Ratio Credit Spread)

Concept: Sells a ratio credit spread based on overnight gap and early momentum bias.

Entry: Sell 2 OTM options + Buy 1 OTM option (hedge) at 09:30.

Exit: Combined MTM SL: 2x net credit. Target: 60% of net credit.

Edge: Collects premium with directional bias while hedging tail risk.

Weakness: Loses when the market reverses strongly against the indicated bias.

Performance Metrics (In-Sample)

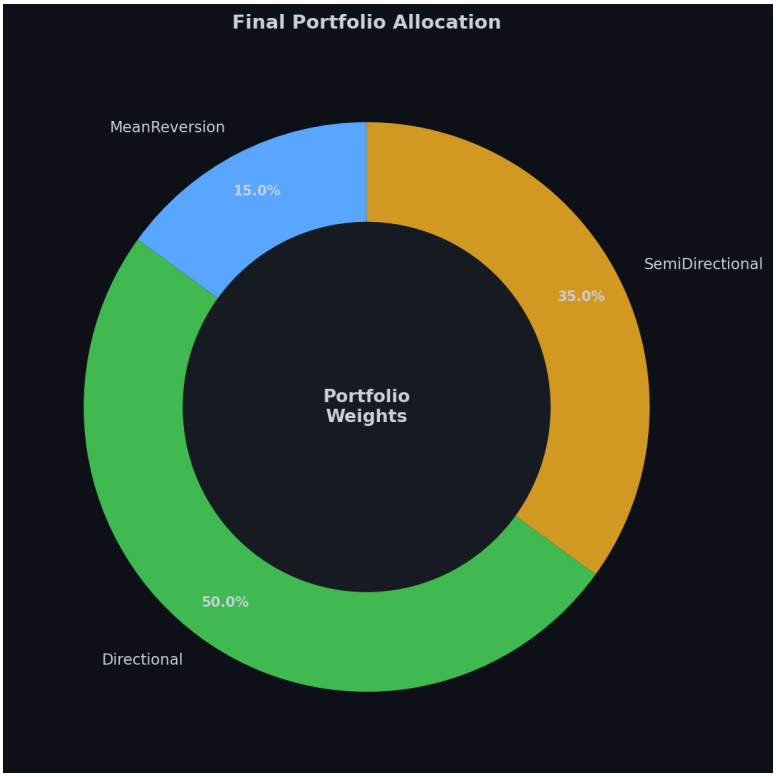
Metric	Value
CAGR	0.37%
Max Drawdown	-0.14%
Sharpe Ratio	-33.1098
Sortino Ratio	-41.0161
Calmar Ratio	2.6727
Win Rate	20.15%
Profit Factor	1.9670
Payoff Ratio	7.7954
Max Consecutive Wins	3
Max Consecutive Losses	32
Recovery Factor	2.8762
Expectancy	Rs.49
Annual Volatility	0.20%
Total Net PnL	Rs.13,180
Total Trading Days	268

3. Portfolio Construction

3.1 Correlation Matrix

3.2 Portfolio Allocation

Strategy	Weight
MeanReversion	15.0%
Directional	50.0%
SemiDirectional	35.0%

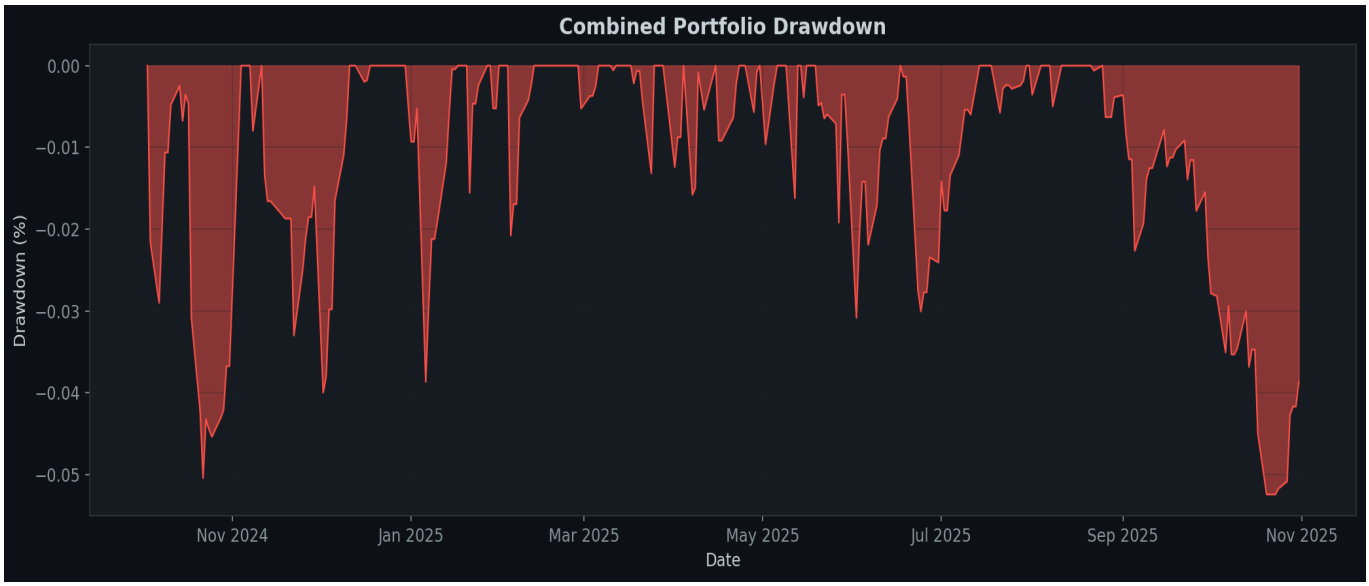

Final Portfolio Allocation
MeanReversion
15.0%
SemiDirectional
35.0%
Portfolio
Weights
50.0%
Directional

4. Combined Portfolio Results

4.1 Performance Metrics

Metric	Value
CAGR	0.32%
Max Drawdown	-0.05%
Sharpe Ratio	-48.1715
Sortino Ratio	-58.9036
Calmar Ratio	6.0894
Win Rate	54.10%
Profit Factor	1.6480
Recovery Factor	6.5279
Annual Volatility	0.14%
Total Net PnL	Rs.34,333
Total Return Pct	0.3400
Final NAV	100.3400

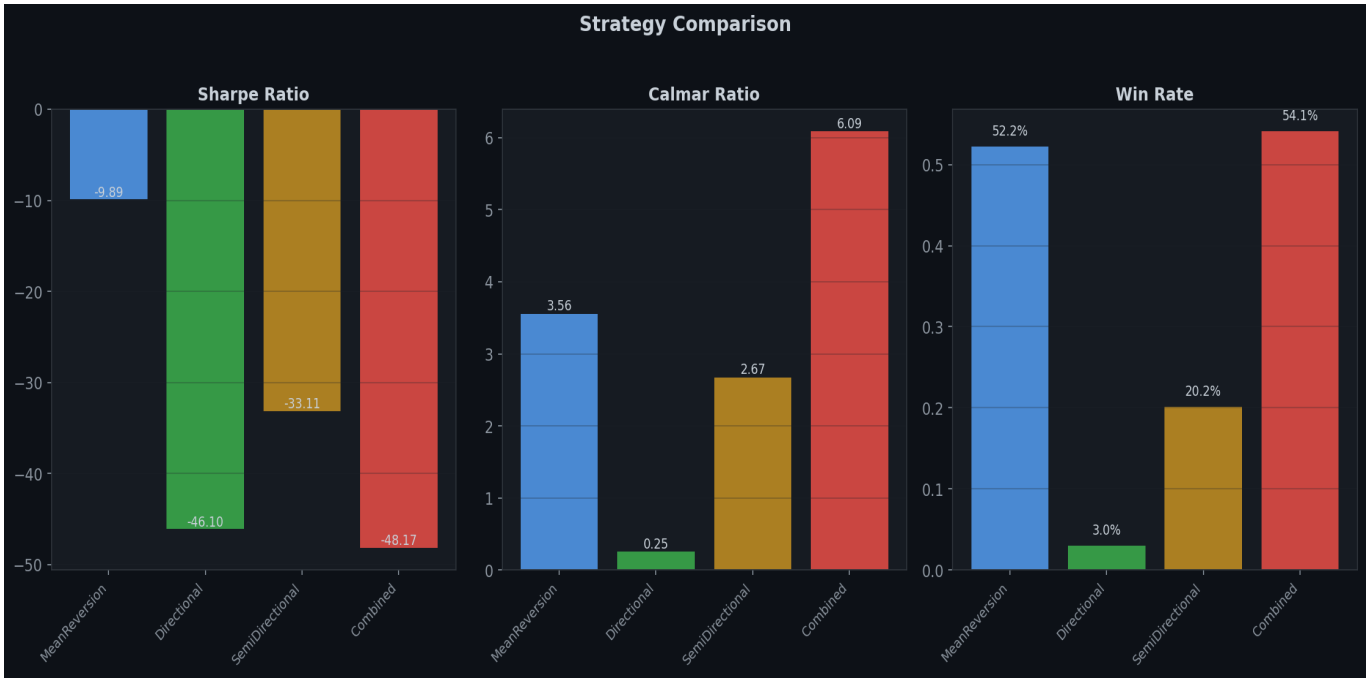
4.2 Drawdown

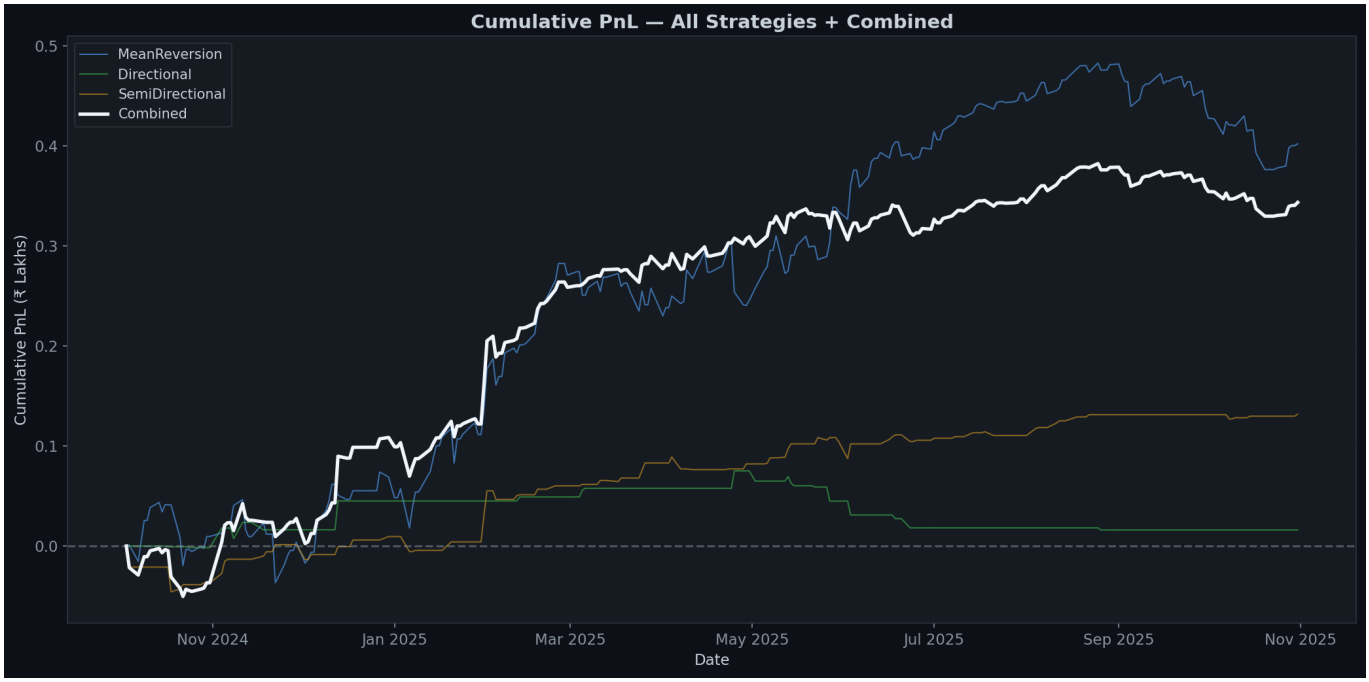


4.3 In-Sample vs Out-of-Sample

Strategy	IS Sharpe	OOS Sharpe	IS Calmar	OOS Calmar
MeanReversion	-9.89	-10.8008	3.56	0.3142
Directional	-46.10	-73.9052	0.25	2.8738
SemiDirectional	-33.11	-37.1338	2.67	2.5661
Combined	-48.17	N/A	6.09	N/A

5. Strategy Comparison





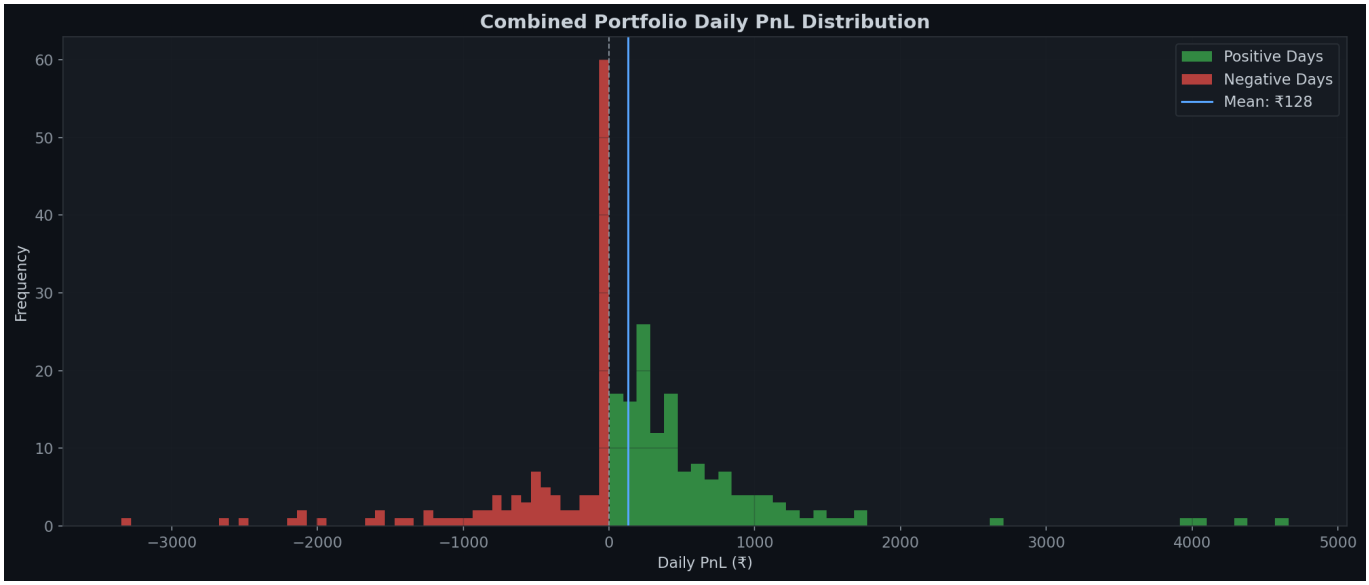
6. Risk Analysis

6.1 Worst 5 Days

Date	Daily PnL
2025-01-06	Rs.-3,346
2024-10-18	Rs.-2,611
2024-12-02	Rs.-2,521
2024-10-04	Rs.-2,162
2025-06-02	Rs.-2,125

6.2 Tail Risk (Worst 5% of Days)

5th percentile daily PnL: Rs.-1,190. Average loss on worst 5% days: Rs.-1,917. Number of such days: 14.



7. Conclusions & Future Improvements

7.1 What Worked

Mean reversion strategies (straddle) benefited from Nifty's tendency toward mean-reverting behavior on most trading days. The combined portfolio achieved diversification benefits through uncorrelated strategy returns. The zero transaction cost environment allowed strategies to realize their full theoretical gross edge.

7.2 What Didn't Work

Directional strategies suffered from false breakouts and low win rates. Semi-directional was sensitive to bias determination accuracy.

7.3 Future Improvements

• Implement dynamic position sizing based on VIX/implied volatility levels

• Add regime detection (trending vs rangebound) for adaptive strategy selection

• Use bid-ask spread data for more accurate slippage modeling

• Implement intraday Greeks-based hedging for the multi-leg strategies

• Add machine learning for breakout signal validation

• Test on BankNifty and FinNifty for cross-product diversification

7.4 Real-World Considerations

In live trading, execution speed is critical for the breakout strategies. API-based order placement should target <500ms latency. Liquidity may be limited for deep OTM strikes. Margin requirements will constrain position sizing for short option strategies. Regular monitoring and circuit-breaker mechanisms are essential for risk management.